Statistical Thinking for Data Science

Lecture 5: Estimation and Optimisation Methods



Lecture 5: Estimation and Optimization Methods

Learning Objectives

- Maximum Likelihood Estimation
- Bayesian method

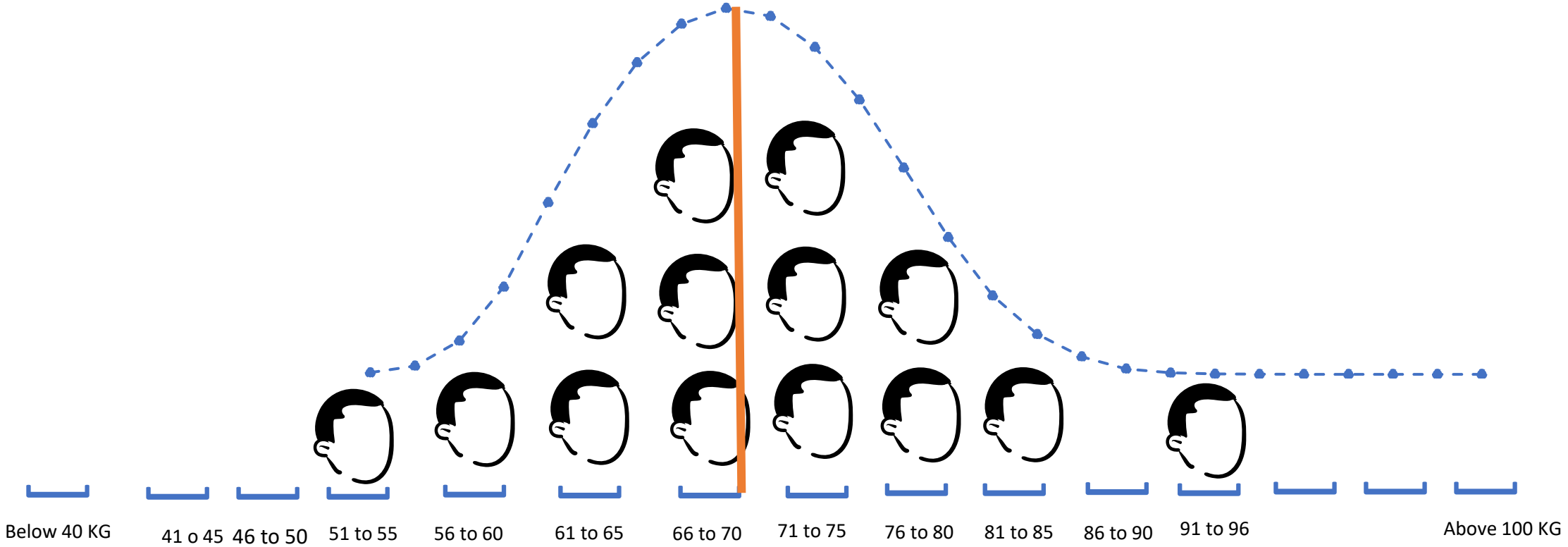
Maximum likelihood Estimation

- Maximum likelihood estimation (MLE) is a method for estimating the parameters of an assumed probability distribution, given some observed data.

- The goal for MLE is to find an optimal way to fit a distribution to the data.

Maximum likelihood Estimation

- The Distribution of Log-Transformed Weights: Normal Distribution



Lecture 5: Estimation and Optimization Methods

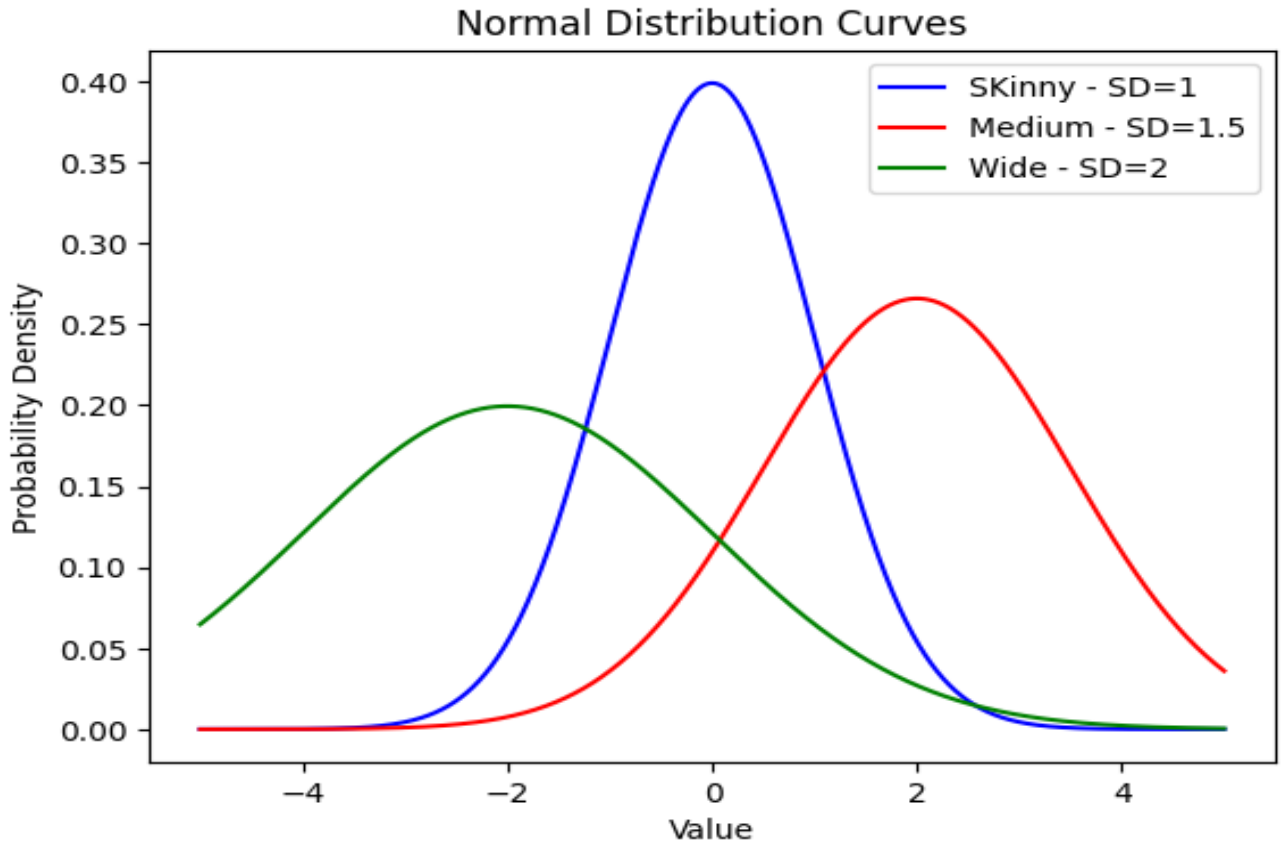
The normal distribution comes in all kinds of shapes and sizes.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$$

$f(x)$ = probability density function

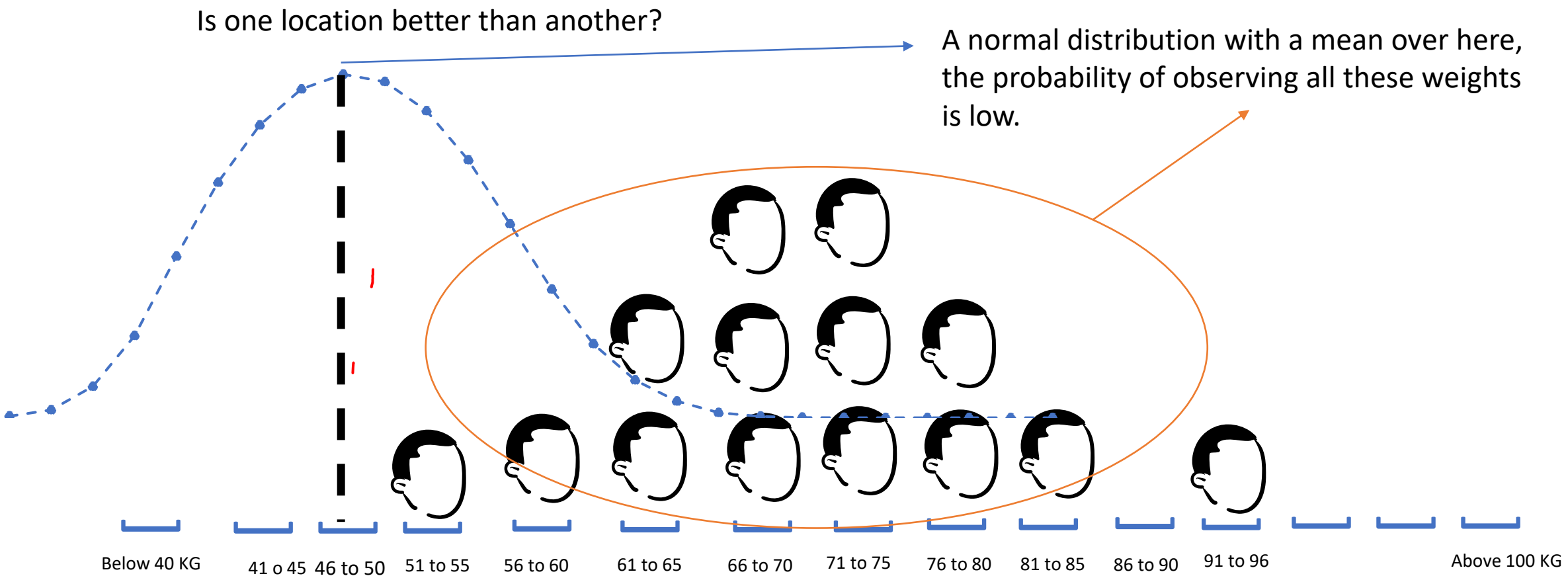
σ = standard deviation

μ = mean



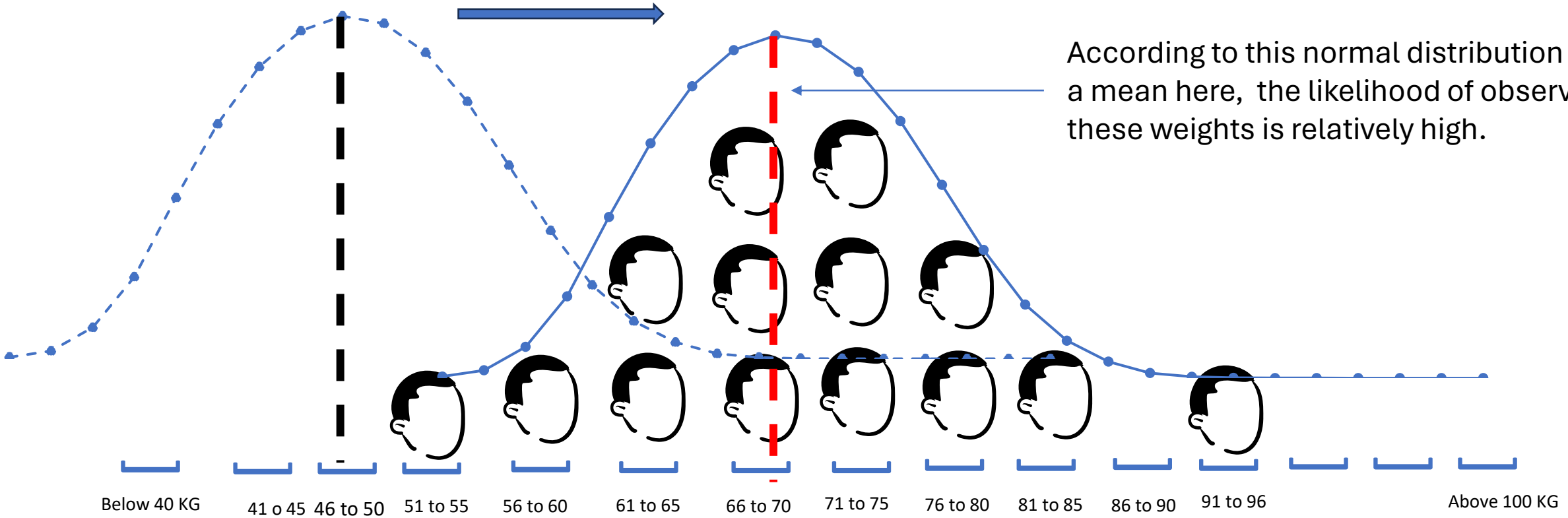
[Normal Distribution at StatDist.com]

Maximum likelihood Estimation



Maximum likelihood Estimation

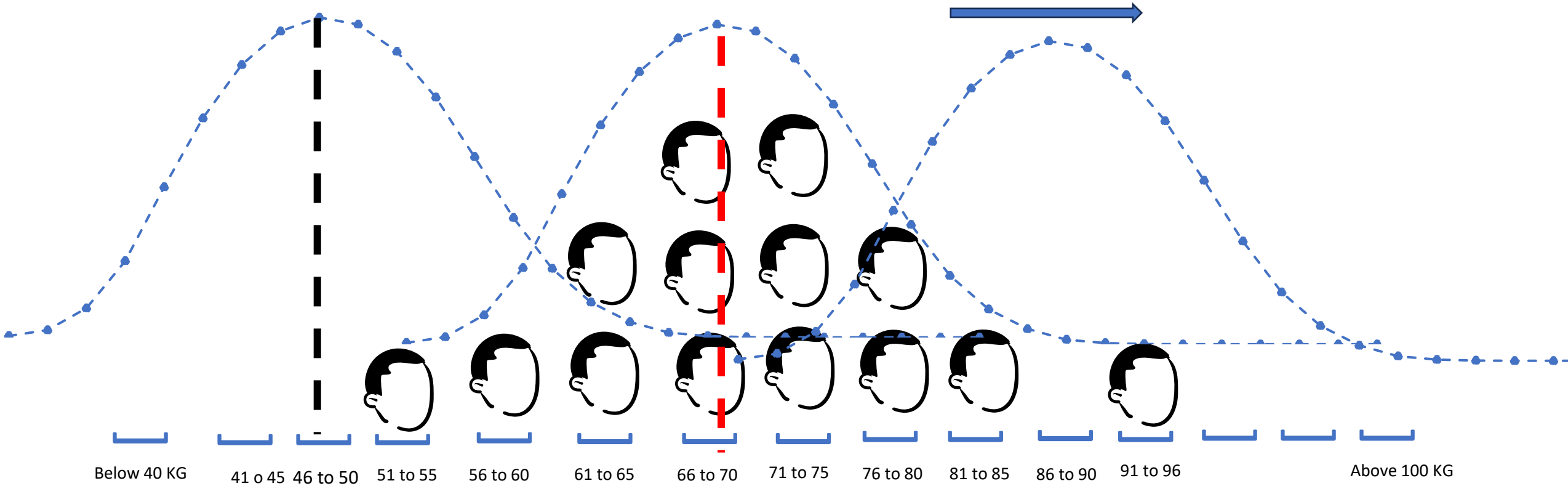
What if we shift the distribution over?



According to this normal distribution with a mean here, the likelihood of observing these weights is relatively high.

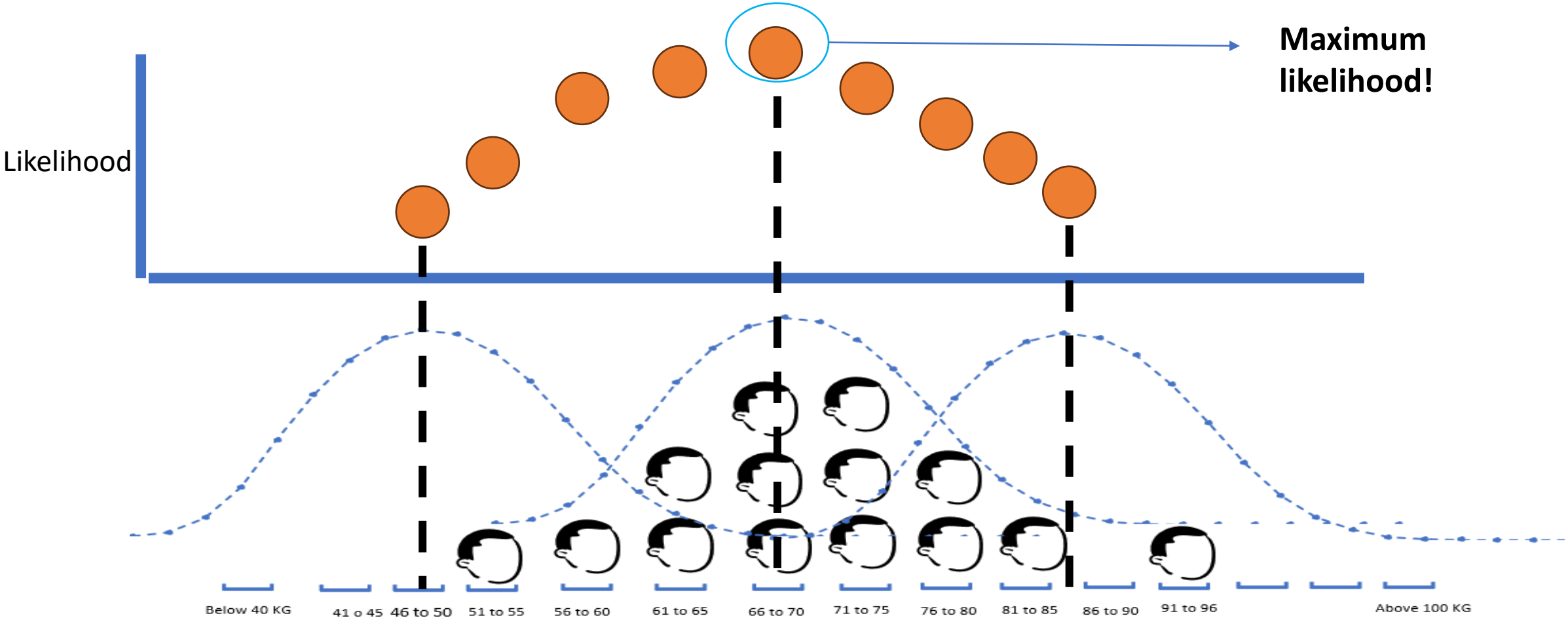
Maximum likelihood Estimation

What if we shift the distribution over?



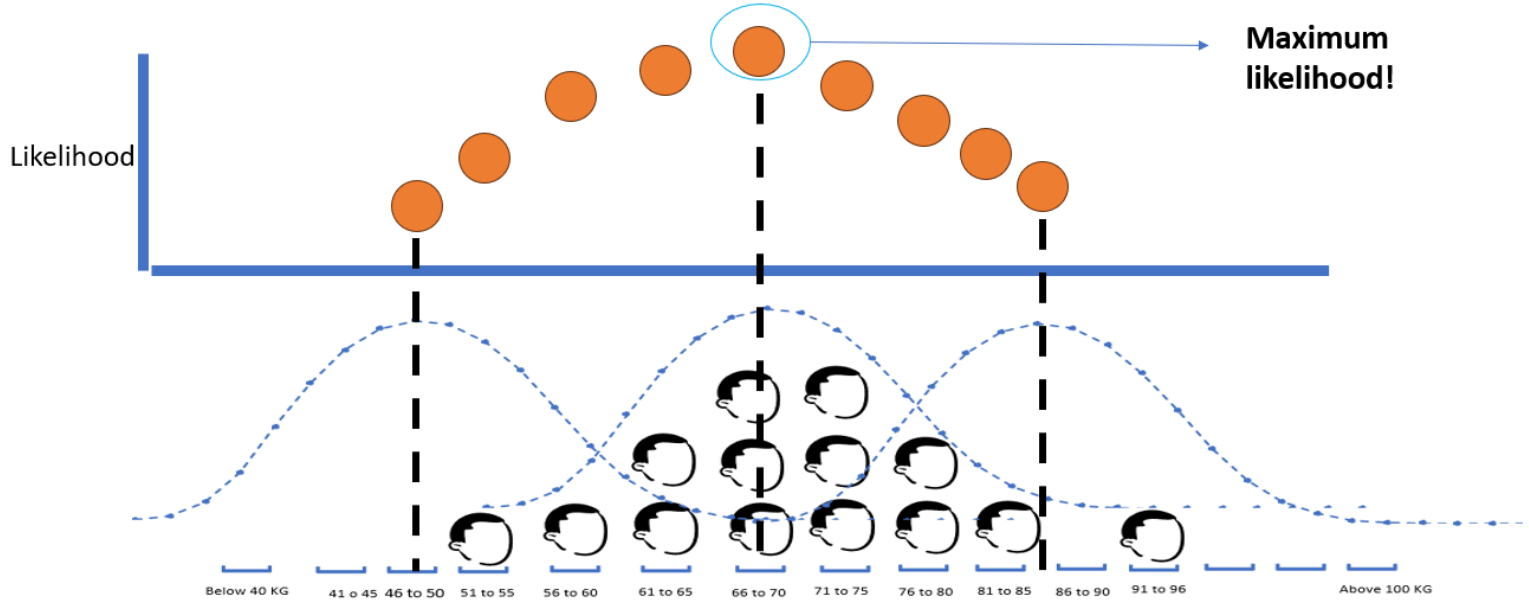
Maximum likelihood Estimation

Where is the best location? We need to calculate the likelihood of observing the data.



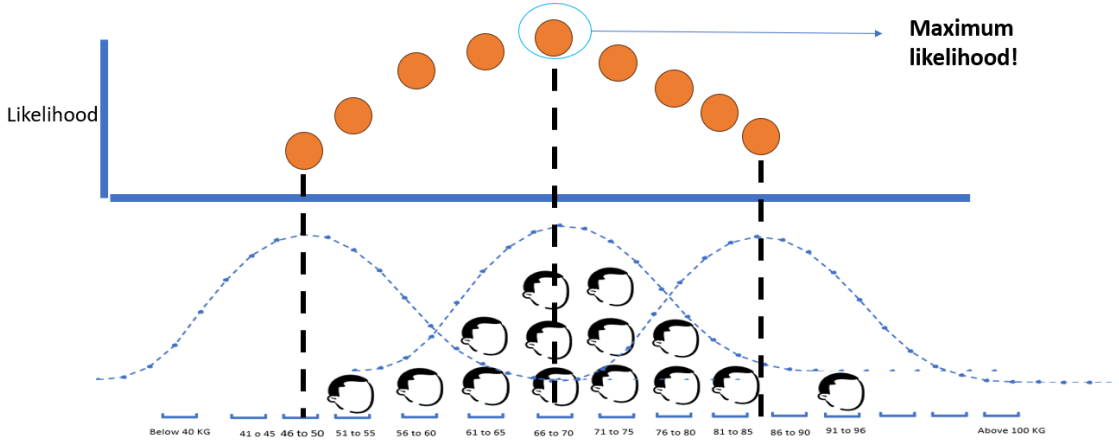
Maximum likelihood Estimation

Question: Why do we need to estimate the location of the centre of the distribution?



Maximum likelihood Estimation

Question: Why do we need to estimate the location of the centre of the distribution?



The location of the centre of the distribution

= mean of the data

= $E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$

Parameter Estimation!

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Likelihood Function

- Definition 5.1 (**Likelihood function**). Let $y_1, y_2, \ldots, y_n$ be a random sample of size n from a discrete pmf or a continuous pdf $f_Y(y; \theta)$, where θ is an unknown parameter. The likelihood function $L(\theta)$ is proportional to the joint pdf evaluated at $y_1, y_2, \ldots, y_n$

$$L(\theta) \propto \prod_{i=1}^{n} f_Y(y_i; \theta). \qquad (1)$$

- In other words, the likelihood is the product of the pmf or pdf evaluated for each data point. It's a function of the parameters.

Likelihood Function Example

Suppose that X is a discrete random variable with the following probability mass function: where $0 \leq \theta \leq 1$ is a parameter.

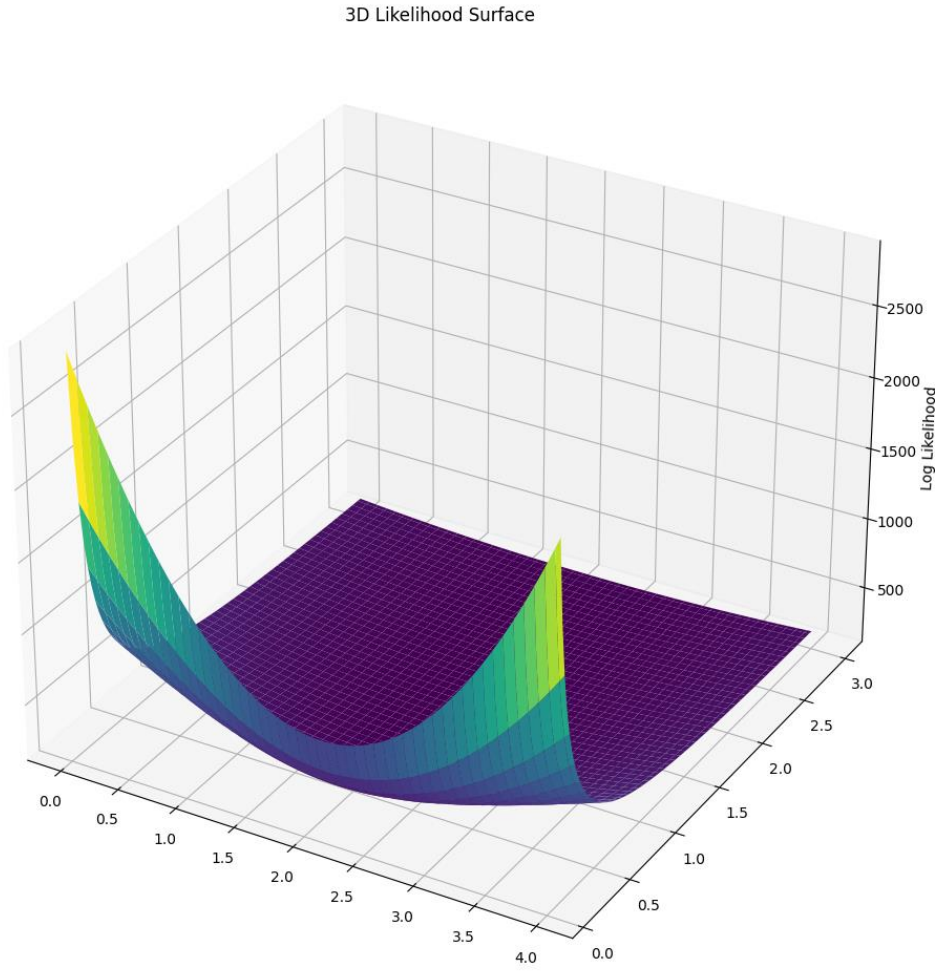
X	0	1	2	3
$P(X)$	$2\theta/3$	$\theta/3$	$2(1-\theta)/3$	$(1-\theta)/3$

The ***likelihood function*** of the sample is (3,0,2,1,3,2,1,0,2,1) is

$$\begin{aligned} L(\theta) = \quad & P(X=3)P(X=0)P(X=2)P(X=1)P(X=3) \\ \times \quad & P(X=2)P(X=1)P(X=0)P(X=2)P(X=1) \end{aligned}$$

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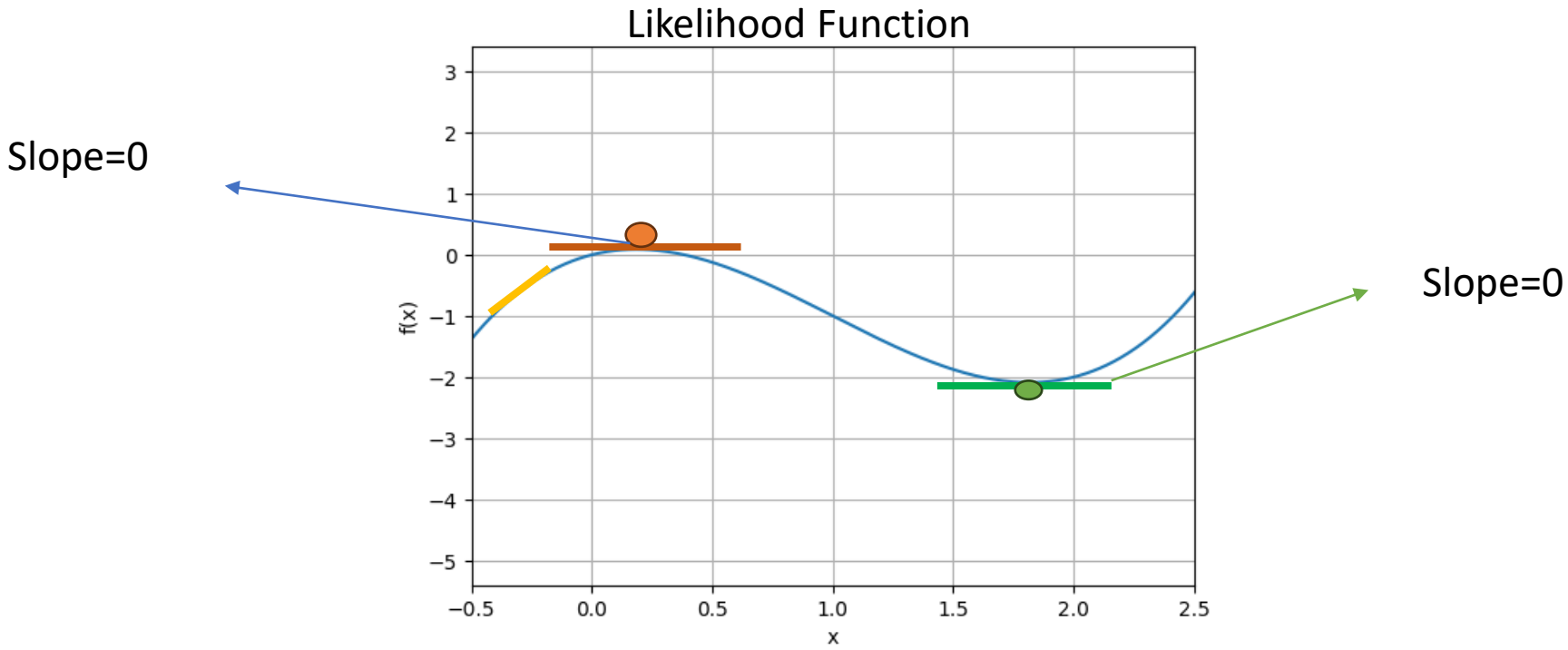
Likelihood Surface



Lecture 5: Estimation and Optimization Methods

Maximum likelihood Estimator

- Finding the maximum likelihood estimator is often an application of Calculus: taking the derivative and setting it equal to zero. It is known as the ***first-order condition.***



Maximum likelihood Estimator

- The first derivative of a function describes the gradient, also known as the slope.

- When we are trying to find the maximum or minimum of a function, we are trying to find the point where the gradient changes from positive to negative or the other way around. When this occurs, the function becomes flat for a moment, and thus the gradient is zero.

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Derivative Example:

- Suppose we have a function $y = x^2 + 3\mathrm{x}$, the slope of this function by definition is

$$\frac{\partial y}{\partial x} = 2\mathrm{x} + 3. \qquad (3)$$

- By setting the slope to zero, we have $x = -1.5$ at the Minium.

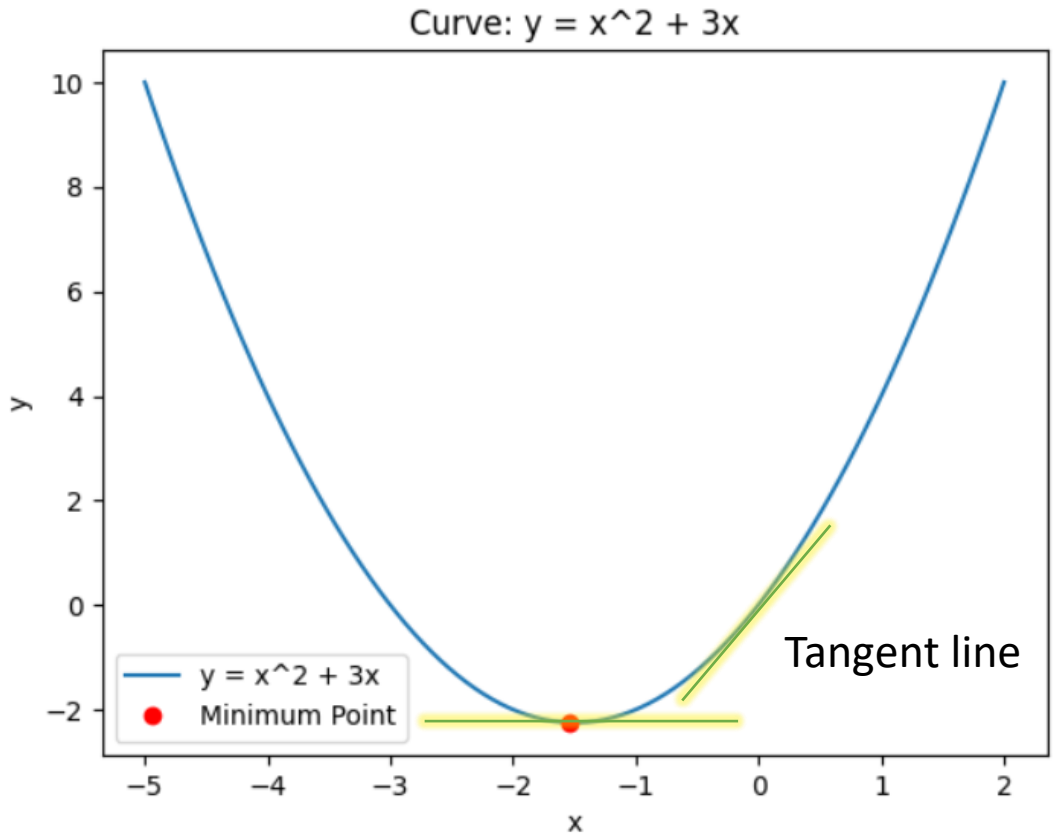


Figure 1: Curve

Critical point

- A critical point of a function f is a point y that has a derivative of zero, $f'(y) = 0$.

- The first step is to try to find the critical points of the function as minima can exclusively arise at these locations.

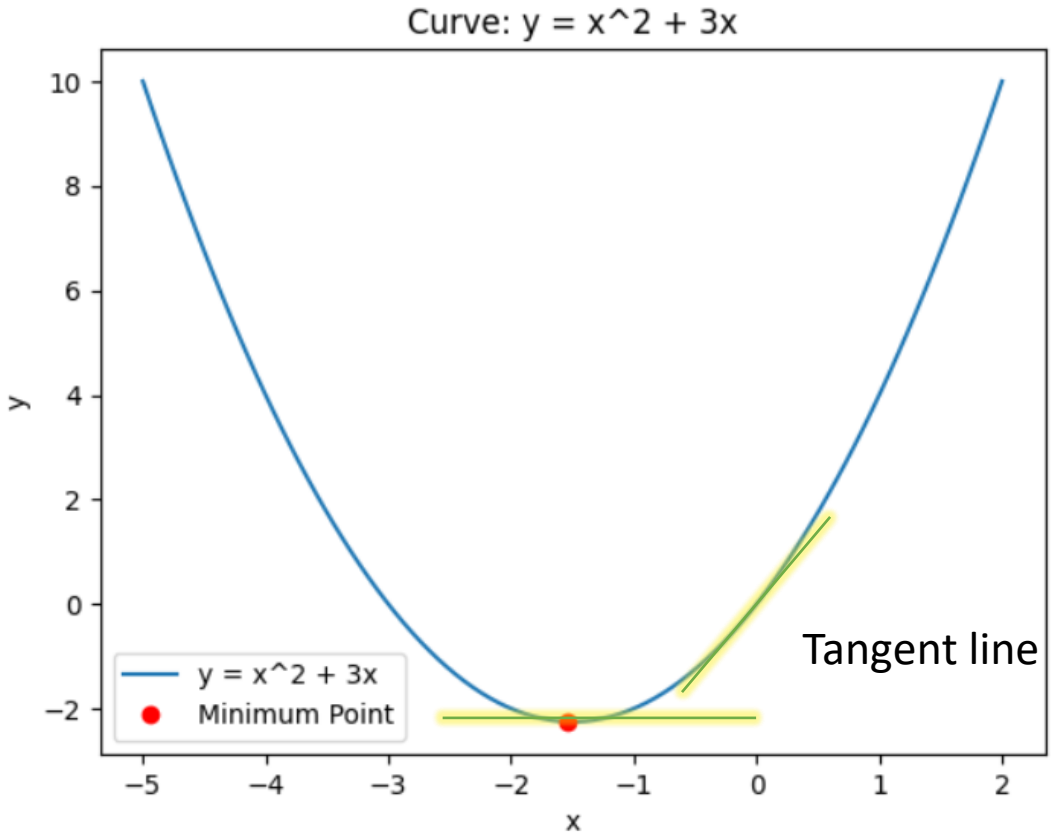
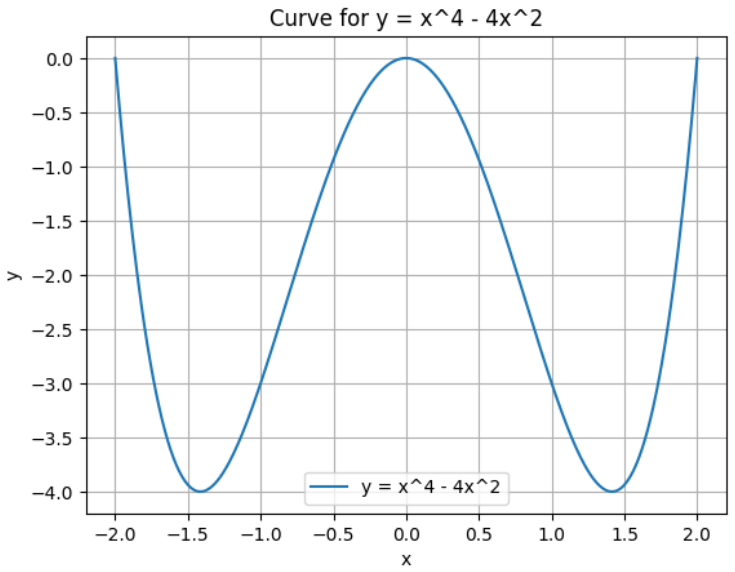


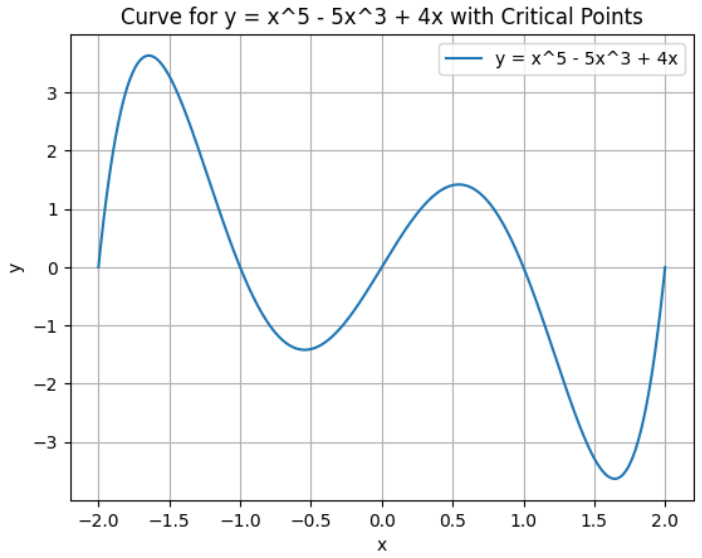
Figure 2: Curve

Lecture 5: Estimation and Optimization Methods

Question:

- Where are the critical points in these graphs?





Critical point

- Critical points serve as key reference points in the optimization process of MLE.
- Critical points encompass various possibilities, including local minimum and maximum points as well as regions indicating flatness.

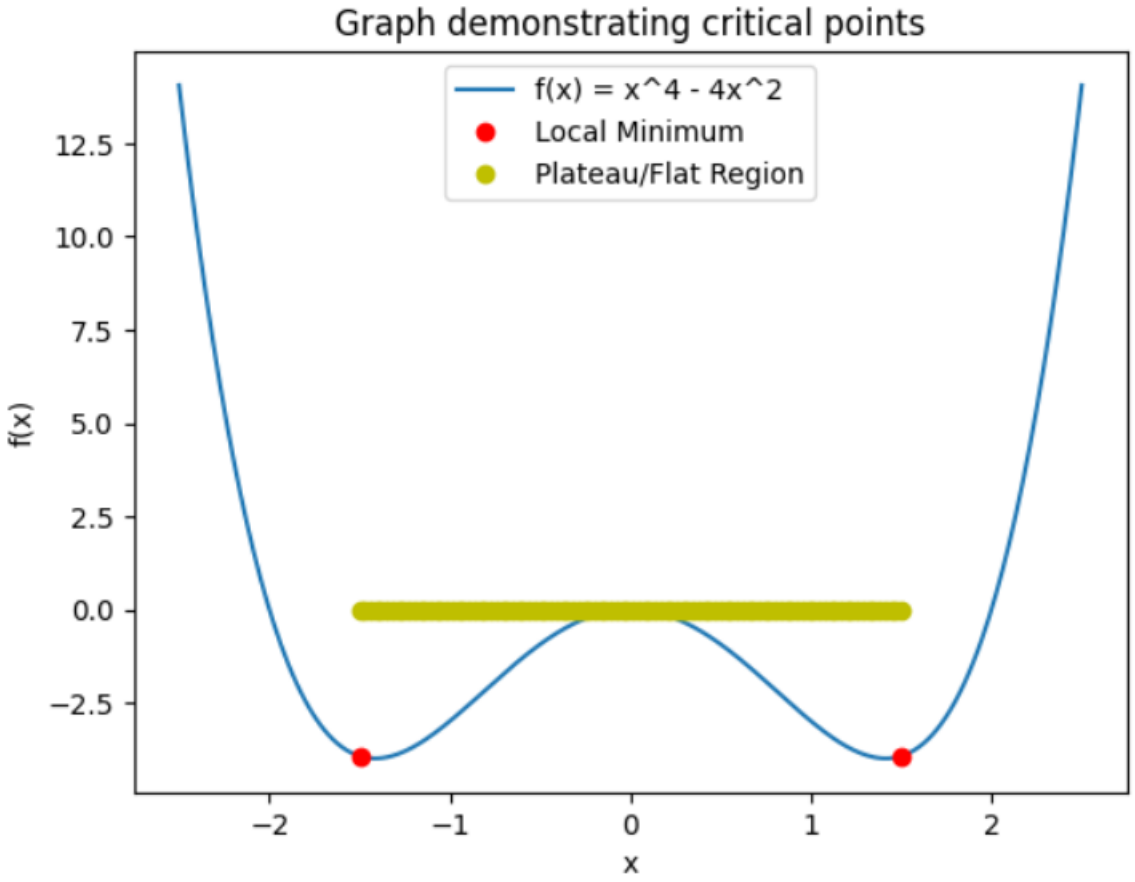


Figure 3: Curve

Local minimum

A local minimum must satisfy the following conditions:

- ***The first order necessary condition***:

$$\text{The first derivative of a function } f'(y;\theta) = 0 \qquad (4)$$

- ***The second order sufficient condition***:

$$\text{The second derivative of a function } f''(y;\theta) > 0 \qquad (5)$$

Convex functions

- If the second derivative of a function $f''(y;\theta) > 0$ for all x ∈ X , then f is strictly convex.

- A function is said to be convex at an interval if, for all pairs of points on the graph, the line segment that connects these two points passes above the curve.

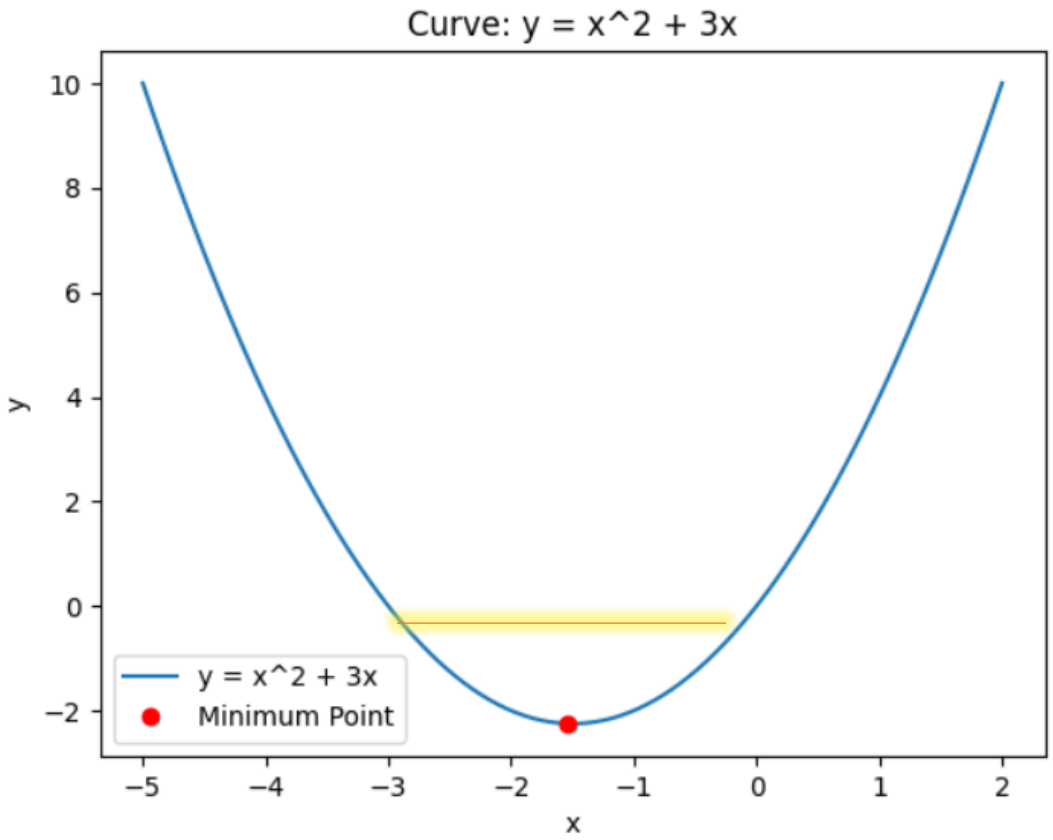


Figure 4: Curve

Convex functions

- Convex functions possess the desirable property that a local minimum automatically qualifies as a ***global minimum.***

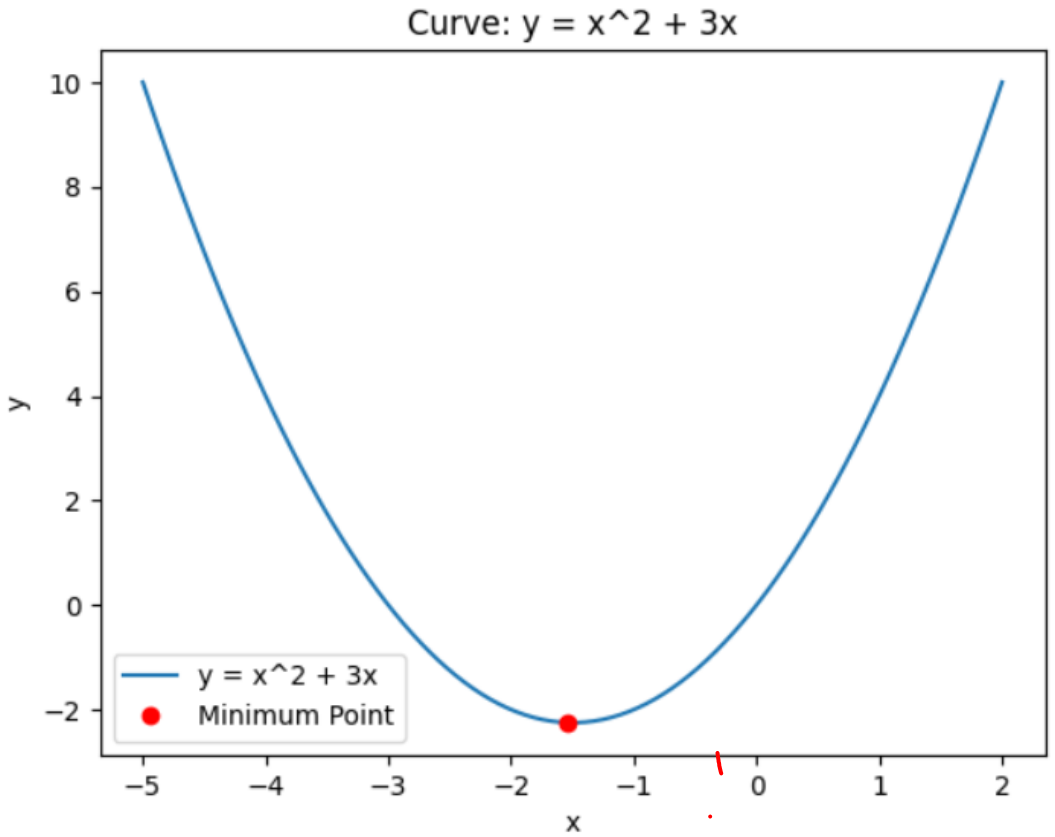


Figure 5: Curve

Concave down functions

- If the second derivative is negative at a point, the graph is concave down.

- If the second derivative is negative at a critical point, then the critical point is a local maximum.

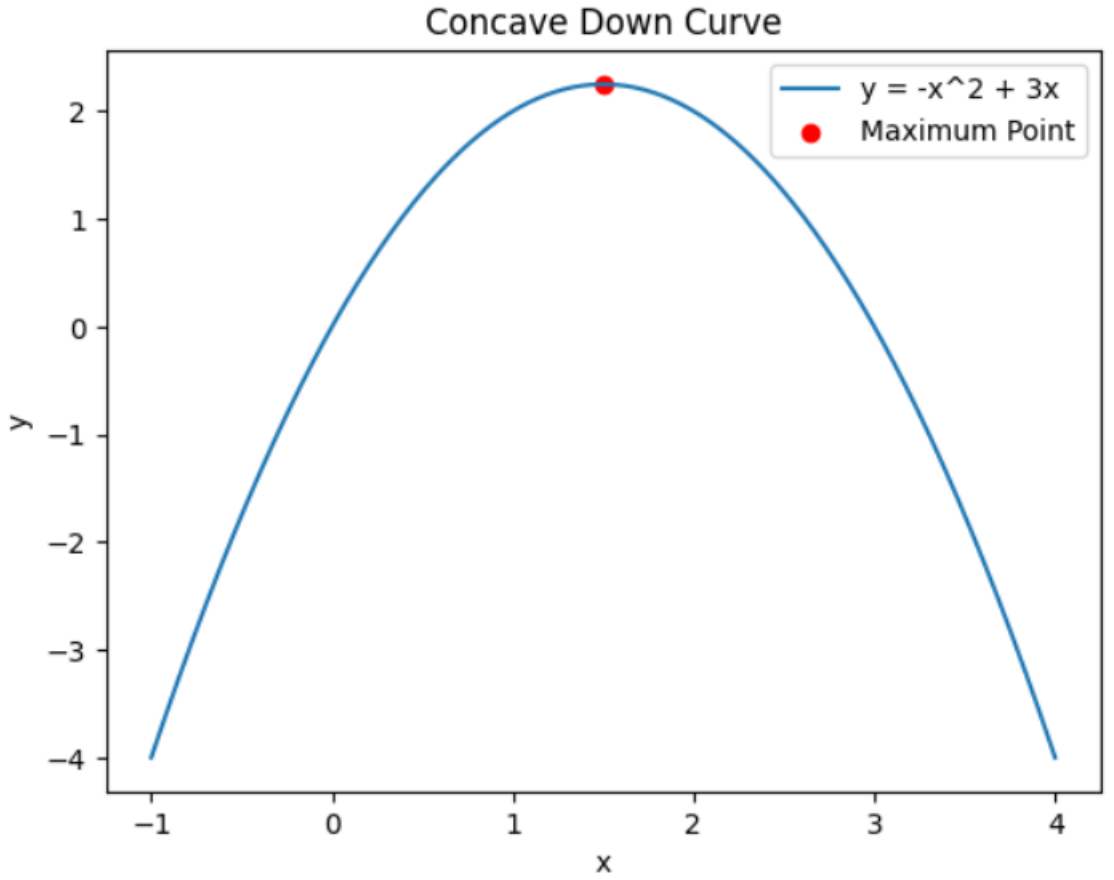
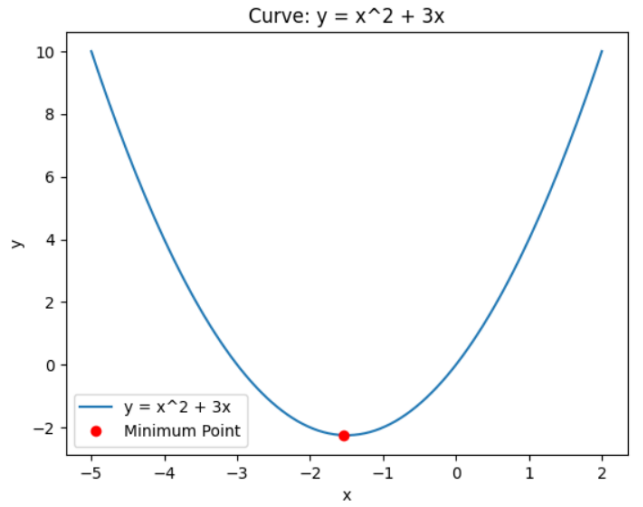


Figure 6: Curve

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Concavity

- **Convex functions :** the second derivative of a function $f''(y;\theta) > 0$
- When the second derivative is zero, it corresponds to a possible inflection point, which is a point on the graph of a function where the curve changes its concavity.
- **Concave down functions:** the second derivative of a function $f''(y;\theta) < 0$. It is commonly referred to as ***the second order condition.***



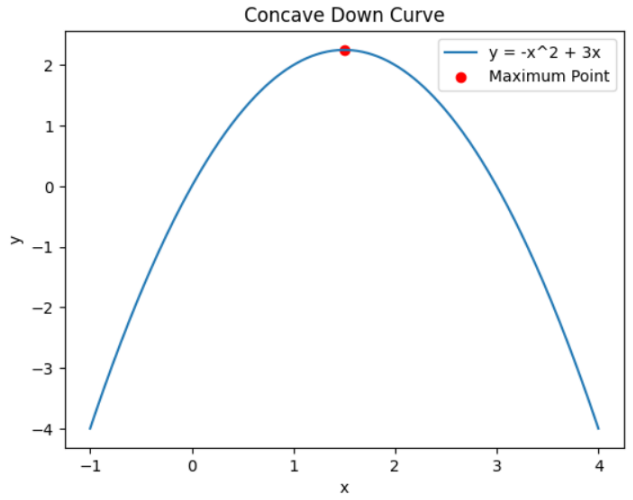


Figure 7: Curve

Maximum likelihood Estimator

- Since the logarithm is a monotonic function, maximizing the $Log(L(\theta))$ or $L(\theta)$ results in the same θ_e .

- The log-likelihood is

$$\ell(\theta) = Log(L(\theta)) \quad (6)$$

Maximum likelihood Estimator

- The negative log-likelihood is

$$\mathcal{L}(\theta) = -\ell(\theta) = -Log(L(\theta)) \quad (7)$$

- By minimizing the negative log-likelihood, we are effectively maximizing the likelihood of observing the given data under a particular model.

Maximum likelihood Estimator Example

- Suppose a bank is evaluating when the customers are more likely to default. The time to default is given by the exponential model with the pdf

$$f_Y(y;\lambda) = \lambda e^{-y\lambda} \qquad y \geq 0. \qquad (8)$$

- Five data points have been collected 9.2; 5.6; 18.4; 12.1; 10.7. Find the maximum likelihood estimate of λ.

Maximum likelihood Estimator Example

- Step 1: Find the likelihood function

$$L(\lambda) \propto \prod_{i=1}^{n} \lambda e^{-\lambda y_i} = \lambda^n e^{-\lambda \sum_{i=1}^{n} y_i}. \qquad (9)$$

- Step 2: Find the log likelihood function

$$\log L(\lambda) = c + n \log \lambda - \lambda \sum_{i=1}^{n} y_i. \qquad (10)$$

Maximum likelihood Estimator Example

- Step 3: Set the first derivative equal to zero, we have

$$\frac{d}{d\lambda}\log L(\lambda) = +\frac{n}{\lambda} - \sum_{i=1}^{n} y_i = 0$$

$$n - \lambda \sum_{i=1}^{n} y_i = 0$$

$$\widehat{\lambda}_{ML} = \frac{n}{\sum_{i=1}^{n} y_i}.$$

Maximum likelihood Estimator Example

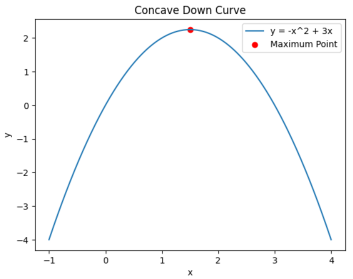
- Step 4: check that this is a point of maximum, we compute the second derivative

$$\frac{d^2}{d\lambda^2} \log L(\lambda) = -\frac{n}{\lambda^2} < 0. \qquad (11)$$

Maximum likelihood Estimator Example

- A negative second derivative indicates that the function is Concave down.

$$\frac{d^2}{d\lambda^2} \log L(\lambda) = -\frac{n}{\lambda^2} < 0. \qquad (13)$$



Maximum likelihood Estimator Example

- Step 5: The final step is to evaluate numerically the formula for $\hat{\lambda}_{ML}$

$$\widehat{\lambda}_{ML} = \frac{5}{9.2 + 5.6 + 18.4 + 12.1 + 10.7} = \frac{5}{56} = 0.089.$$

Maximum likelihood Estimation Example

- Maximum Likelihood (ML) is a widely adopted approach for estimating regression coefficients in various applications.
- Logistic regression example
- Consider a scenario where a bank seeks to predict the probability of a customer defaulting. To simplify the analysis, let's focus on a single variable: income.

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Maximum likelihood Estimation Example

Logistic regression example

- We consider using a logistic regression model to predict the probability of default (PD):

$$p(X) = \frac{e^{\beta_0+\beta_1 X}}{1+e^{\beta_0+\beta_1 X}}. \qquad (13)$$

Where $p(x)$ $is\ the\ problility\ of\ default,$ β_0 and β_1 denote the coefficients to be estimated and X is the variable of income.

Lecture 5: Estimation and Optimization Methods

ML Example – logistic regression parameter estimation

- Bernoulli distribution is a discrete distribution of a random variable which takes the value 1 with probability p and the value 0 with the probability $1 - p$.
- The response variable default or not $Y_i|X_i = \boldsymbol{x}_i \sim \text{Bernoulli}\left(\pi(\boldsymbol{x}_i)\right)$,
- The probability mass function (pmf) is

$$p_{Y|X}(y_i|\boldsymbol{x}_i) = \pi_i^{y_i}(1-\pi_i)^{1-y_i}, \qquad (14)$$

ML Example – logistic regression parameter estimation

- So far, we define

$$f(x_i)=\beta_0+\beta_1 x_i \quad (15)$$

$$\sigma(z) = \frac{1}{1+\exp(-z)} \quad (16)$$

$$\pi(x_i) = \sigma(f(x_i)) = \sigma(\beta_0+\beta_1 x_i) \quad (17)$$

Lecture 5: Estimation and Optimization Methods

ML Example – logistic regression parameter estimation

- The log-likelihood function is

$$
\begin{aligned}
\ell(\boldsymbol{\beta}) &= \log\left(\prod_{i=1}^{n} \pi_i^{y_i}(1-\pi_i)^{1-y_i}\right) \\
&= \sum_{i=1}^{n} y_i \log(\pi_i) + (1-y_i)\log(1-\pi_i) \\
&= \sum_{i=1}^{n} y_i \log\left(\sigma\left(\beta_0 + \sum_{j=1}^{p}\beta_j x_{ij}\right)\right) \\
&\quad + (1-y_i)\log\left(1-\sigma\left(\beta_0 + \sum_{j=1}^{p}\beta_j x_{ij}\right)\right)
\end{aligned}
$$

ML Example – logistic regression parameter estimation

- The negative likelihood function is

$$\mathcal{L}(\beta) = -\ell(\beta) \tag{18}$$

- Our objective is to estimate $\boldsymbol{\beta}$
- By minimizing the negative log-likelihood, we are effectively maximizing the likelihood of observing the given data.

Lecture 5: Estimation and Optimization Methods

ML Example – logistic regression parameter estimation

- The first order condition : setting the first derivative of the negative log likelihood to zero.

$$-\frac{\sum y_i}{\widehat{\pi}} + \frac{n - \sum y_i}{1 - \widehat{\pi}} = 0,$$

- Solving for $\hat{\pi}$, we find the critical point

$$\widehat{\pi} = \frac{\sum_{i=1}^{n} y_i}{n}.$$

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ML Example – logistic regression parameter estimation

- The second order condition : obtain the second derivative of the negative log likelihood function.

$$J''(\pi) = \frac{\sum y_i}{\pi^2} + \frac{n - \sum y_i}{(1-\pi)^2}, \qquad (19)$$

- Since $\pi \in (0,1)$, J'' is positive and therefore J'' is a convex function.
- As such, $\hat{\pi}$ is the global minimum.

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ML Example – logistic regression parameter estimation

- Estimating the coefficients $\boldsymbol{\beta}$ or $\boldsymbol{\theta}$

$$\frac{d}{d\theta_1} \log L(\theta_1, \theta_2, \ldots, \theta_k) = 0$$
$$\frac{d}{d\theta_2} \log L(\theta_1, \theta_2, \ldots, \theta_k) = 0$$
$$\vdots$$
$$\frac{d}{d\theta_k} \log L(\theta_1, \theta_2, \ldots, \theta_k) = 0$$

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ML Example – logistic regression parameter estimation

- Estimating coefficients example
- Suppose the time to failure of devices follows an exponential distribution, which is a special case of the Weibull distribution. The probability of failure increases as time goes. The Weibull random variable Y has pdf

$$f_Y(y;\alpha,\beta) = \alpha\beta y^{\beta-1}e^{-\alpha y^{\beta}}, \quad 0 < y, 0 < \alpha, 0 < \beta. \tag{20}$$

Lecture 5: Estimation and Optimization Methods

ML Example – logistic regression parameter estimation

- Our objective to is to estimate coefficients.
- Step 1: Derive the likelihood function

$$L(\alpha,\beta)=\prod_{i=1}^{n}\alpha\beta y_i^{\beta-1}e^{-\alpha y_i^{\beta}}=\alpha^n\beta^n\left(\prod_{i=1}^{n}y_i\right)^{\beta-1}e^{-\alpha\sum_{i=1}^{n}y_i^{\beta}}, \tag{21}$$

- Step 2: write down the log likelihood function

$$\log L(\alpha,\beta)=n\log\alpha+n\log\beta+(\beta-1)\log\left(\prod_{i=1}^{n}y_i\right)-\alpha\sum_{i=1}^{n}y_i^{\beta}. \tag{22}$$

Lecture 5: Estimation and Optimization Methods

ML Example – logistic regression parameter estimation

- Our objective to is to estimate coefficients.
- Step 3: differentiate the log likelihood function with respect to each parameter

$$\frac{d \log L(\alpha, \beta)}{d\alpha} = \frac{n}{\alpha} - \sum_{i=1}^{n} y_i^{\beta}, \tag{23}$$

$$\frac{d \log L(\alpha, \beta)}{d\beta} = \frac{n}{\beta} + \log \prod_{i=1}^{n} y_i - \alpha \sum_{i=1}^{n} y_i^{\beta} \log y_i, \tag{24}$$

ML Example – logistic regression parameter estimation

- Step 4: We need to solve these two equations for α and β

$$\begin{cases} \frac{n}{\alpha} - \sum_{i=1}^{n} y_i^{\beta} = 0, & (25) \\ \frac{n}{\beta} + \sum_{i=1}^{n} \log y_i - \alpha \sum_{i=1}^{n} y_i^{\beta} \log y_i = 0. & (26) \end{cases}$$

- Step 5: The second derivative is

$$\frac{d^2 \log L}{d\alpha^2} = -\frac{n}{\alpha^2} < 0 \qquad (27)$$

Therefore, the estimated parameters are MLE.

Maximum Likelihood Applications

- **Regression Analysis**: MLE is used to estimate the parameters of regression models and GLMs, such as logistic regression, and Poisson regression.

- **Machine Learning and Advanced Models:** In machine learning, MLE is used for training various models, such as maximum likelihood estimation for the parameters of a Mixture Model.

Maximum Likelihood Applications

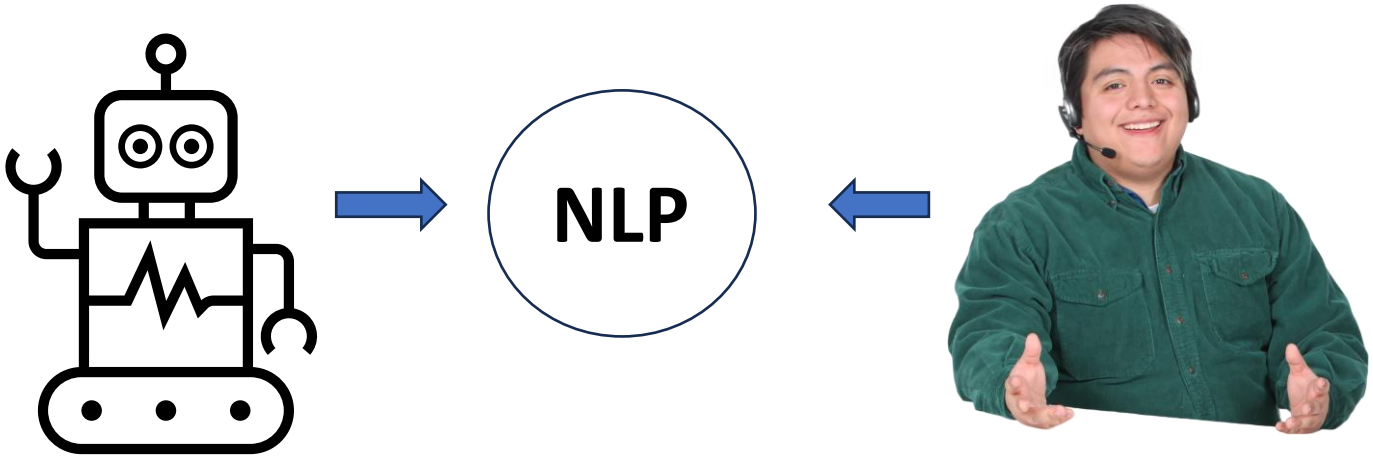
- **Image Processing:** MLE can be applied in image processing for tasks such as image denoising where it estimates parameters based on observed data and underlying models.

- **Natural Language Processing (NLP):** MLE is used for estimating probabilities in language models, where it calculates the likelihood of a word given its context.

Maximum Likelihood Applications

Natural Language Processing

Natural Language Processing (NLP) is a sub-field of artificial intelligence (AI), dealing with processing and generating natural language data.



Natural Language Processing

NLP: Sequential Data

- The text that appears in a sentence is sequence data.
- Goal: Predict P(Word | History)
- Example: P(“Autumn ”|“I am so excited to be taking 36103 this”) > P(“Coffee ”|“I am so excited to be taking 36103 this”)

Natural Language Processing

- How do we predict these probabilities?
  - Count the number of times you see the history
  - Count the number of times you observed the specified word following the history

    Example: P(“Autumn ”|“I am so excited to be taking 36103 this”)

    = C (“I am so excited to be taking 36103 this autumn”) /C (“I am so excited to be taking 36103 this”)

Natural Language Processing

- Maximum Likelihood Estimation in NLP
  - If we want to find the probabilities of entire sequences, we can just multiply these individual word probabilities to get the probability of a word sequence.

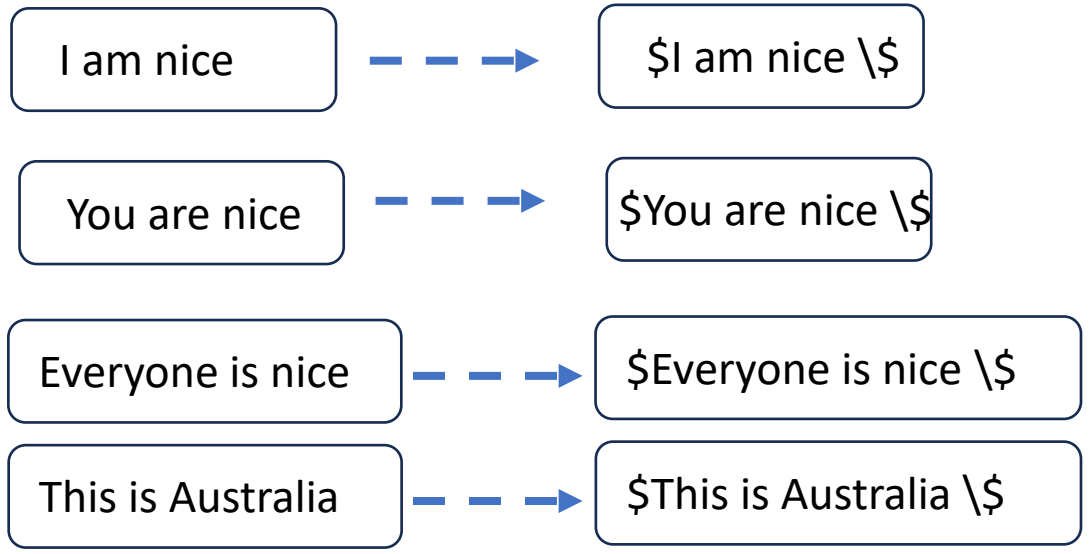
$$P(w_1^n) \approx \prod_{k=1}^{n} P(w_k | w_{k-N+1}^{k-1})$$

  - Examples: P(“Winter break is coming soon”)

= P(“soon”|“coming”) * P(“coming”|“is”) * P(“is”|“break”) *P(“break”|“Winter”)

Natural Language Processing

- Examples of Maximum Likelihood Estimation in NLP



Words	Frequency
$ I	1
I am	1
am nice	1
Nice \$	3
….	
Australia \$	1

Natural Language Processing

Examples of Maximum Likelihood Estimation in NLP

Words	Frequency	Single Word	Frequency
$ I	1	$	4
I am	1	am	1
am nice	1	are	1
Nice \$	3	Nice \$	3
….		….	
Australia \$	1	Australia \$	1

- P(“I”|“$”) = C(“$”|“I”) /C(“$”) =1/4=0.25
- This means the MLE of a sentence starting with “I” is 25%.

Natural Language Processing

How do computers recognize all these words?

Natural Language Processing involves a series of key steps to process and understand human language. The main steps in NLP are:

1. Text Preprocessing

Tokenization: Splitting text into individual words or tokens.

- Example: Original text “ What a great day today!”
- Tokenization “What”, “a”, “great”, “day”, “today”

Natural Language Processing

2. Word to Vector Transformation :

- Next, we need to construct a vocabulary that will enable us to convert the tokenized words into sequences of token IDs, a format necessary for predictive models like neural networks.

- Examples:
- Tokenization “What”, “a”, “great”, “day”, “today”
- Text transformation [2, 100, 7, 30, 2000]

Natural Language Processing

3. Word embedding

- The hypothesis in linguistics states that words with similar meanings should occur in similar contexts.
- In other words, from a word we can get an idea about the context where it might appear.
- Examples:

__ __ car __ __ **Max L** (Context |Word)

The red __ is nice **Max L** (Word |Context)

Natural Language Processing

3. Word embedding

- Word Embeddings (e.g., Word2Vec) are pre-trained models that map words to fixed-length vectors where similar words are represented by vectors that are closer in the vector space.
- These embeddings capture ***semantic*** relationships between words, making them suitable for various NLP tasks.
- Examples:
  - The vectors for "cat" and "kitten" will be close.
  - "Big" and "large" will have similar vectors.

Natural Language Processing

4. Language Models

- The aim of a language model is to minimize how confused the model is having seen a given sequence of text.

- MLE is used to estimate the probabilities of words or sequences of words in a given language.

- Examples: suppose we have a simple corpus with the following sentences.

Natural Language Processing

Examples

“ I like to eat pizza.”

“I like to eat pasta.”

“I like to eat burgers.”



- We want to estimate the probability of the word "pizza" appearing after the word "eat." We can use MLE to calculate this probability.
- Probability of “pizza” given “eat” (P(pizza|eat)) = Count(“eat pizza”) / Count(“eat”) = 1/3.

Natural Language Processing

Examples

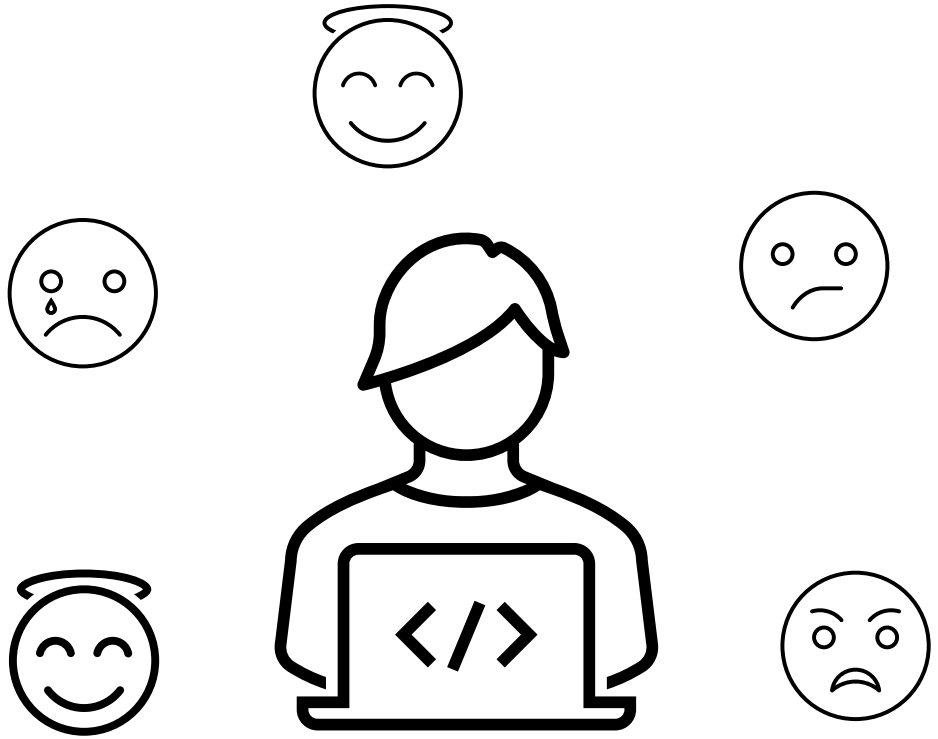
These probabilities are then used in tasks like

- Speech Recognition
- Machine Translation
- Text Generation

Sentiment Analysis

Sentiment Analysis

It is the field that enables computers to understand user emotions.



Sentiment Analysis

Positive and Negative words

- We frequently use words to express our emotions or convey our sentiments about something.
- People often attach connotations or meanings to these words
- Examples:
  - Good, yes, friendship transmits positive feelings
  - Bad, hate and no transmit negative feelings

Sentiment Analysis

Sentiment

- We use words in a text to determine the sentiment behind the text.
- The simplest approach:
- Count the number of Positive P and negative N words

$$\text{Sentiment} = (P-N)/ (P+N)$$

- This effectively assigns a weight of 1 to positive texts and -1 to negative texts and defines how dominant one sentiment is over another.
- Despite its simplicity, it's remarkably powerful.

Recurrent Neural Network

Recurrent Neural Network (RNN)

- RNN is a type of neural network designed to process sequential data, such as time series or natural language.
- ***Recurrent Connections***: RNNs have recurrent connections that allow information to flow from one time step to the next, creating a dynamic relationship between past and current inputs.

Summary

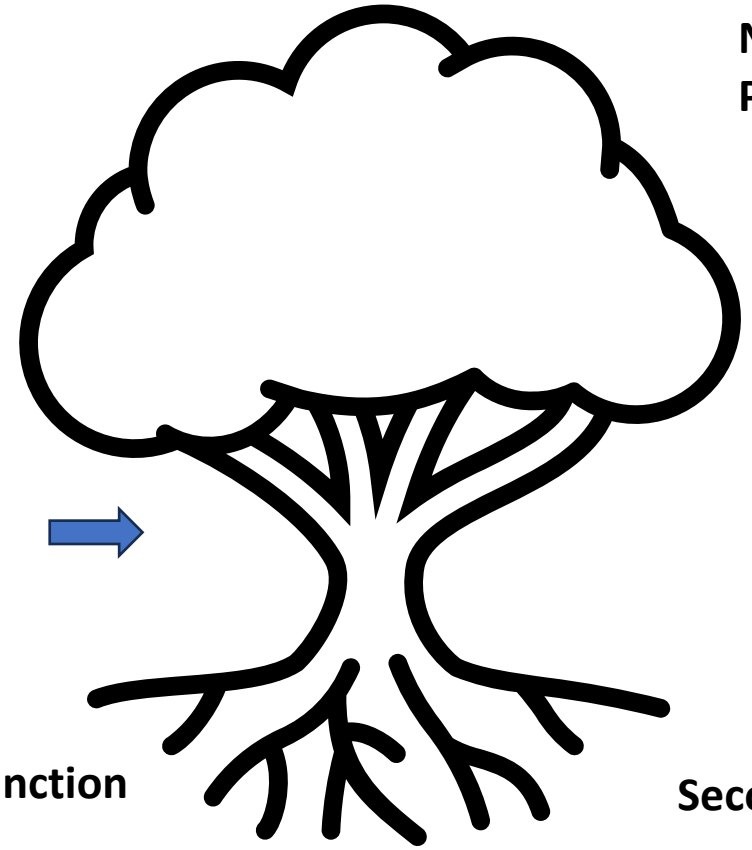
Applications

Parameter Estimation

Computer Vision

Natural Language Processing

Maximum Likelihood Estimation



Foundations

Likelihood Function

First Order Condition

Second Order Condition

References

1. James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021). An Introduction to Statistical Learning with Applications in R (Second Edition). New York: Springer. (An Introduction to Statistical Learning (statlearning.com))

2. Larsen RL, Marx ML (2012). Introduction to Mathematical Statistics and Its Applications, 5th Edition, Boston: Pearson, Section 4 and 5.

3. Kostadinov, Simeon. Recurrent Neural Networks with Python Quick Start Guide : Sequential Learning and Language Modeling with TensorFlow, Packt Publishing, Limited, 2018. ProQuest Ebook Central, https://ebookcentral.proquest.com/lib/uts/detail.action?docID=5609746.